

Project Design Phase-II

Data Flow Diagram & User Stories

Date	June 2026
Project Name	AI-Enhanced Intrusion Detection System
Maximum Marks	As Applicable

Data Flow Diagrams:

A Data Flow Diagram (DFD) shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 (Context Diagram):

External Entity: Security Analyst ↔ System: AI-Enhanced IDS ↔ External Entity: Network Traffic Source

DFD Level 1 (Simplified):

1. Network Traffic → [1.0 Feature Extraction] → Preprocessed Features
2. Preprocessed Features → [2.0 ML Classification] → Attack Class + Confidence
3. Attack Class → [3.0 Flask API] → JSON Response
4. JSON Response → [4.0 Dashboard Rendering] → SOC Analyst View
5. SOC Analyst → [5.0 Log Export] → CSV Audit File

User Stories:

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
SOC Analyst	Real-time monitoring	USN-1	As a SOC analyst, I can view a live dashboard showing current network threat status and alert counts	Dashboard loads with current stats in under 3 seconds	High	Sprint-1
SOC Analyst	Threat classification	USN-2	As a SOC analyst, I can see each detected threat classified as Normal, DoS, Probe, R2L, or U2R with a confidence score	Correct classification displayed with percentage confidence	High	Sprint-1
SOC Analyst	Alert feed	USN-3	As a SOC analyst, I can view a real-time feed of recent threat alerts with timestamp, type, and source IP	Alert feed updates automatically every 5 seconds	High	Sprint-1

SOC Analyst	Attack analytics	USN-4	As a SOC analyst, I can view charts showing attack type distribution and frequency over time	Bar and pie charts render correctly with Chart.js	Medium	Sprint-2
SOC Analyst	API prediction	USN-5	As a developer/analyst, I can send network features to the /predict API endpoint and receive a threat classification	API responds in under 200ms with attack class and confidence	High	Sprint-1
SOC Analyst	Model accuracy	USN-6	As a SOC analyst, I can trust the model maintains at least 95% classification accuracy on the NSL-KDD test set	Model evaluation reports accuracy $\geq 95\%$	High	Sprint-2
SOC Analyst	Log export	USN-7	As a SOC analyst, I can export detected threat logs as a CSV file for audit and compliance purposes	CSV file downloads successfully with all alert data	Low	Sprint-3